

HAFIDI's work uses GA to optimize the hyperparameters of Deep Learning models, specifically RNN, LSTM, Bi-LSTM, and GRU. The data used are cryptocurrencies such as Bitcoin, Ethereum, and Litecoin from 2017 to 2021, focusing less on technical indicators and more on historical prices and volatility. The evaluation metrics were R², MAPE, RMSE, MSE, and MAE. MESKAS's work uses a hybrid GA-LSTM-CNN neural network as its main model, employing GA for feature selection and finding an optimal model structure. It investigates the influence of chart patterns on the forecasting of 20 companies from 2013 to 2024. In addition to charts, it uses technical indicators such as RSI, EMA, and MACD. Its evaluation metrics include RMSE, accuracy, and confusion matrix. Thus, these works share with this article the use of GA to optimize neural network models, although their indicators, apart from the EMA, are different from it; they are momentum indicators, in addition to the use of volatility, techniques that are present in the indicators used in this work.

The GUO work conducts an empirical study on the effectiveness of technical indicators for forecasting S & P 500 stocks, with over 90 years of data. Different machine learning models are used, such as Random Forest, in addition to the LSTM neural network. More than 70 technical indicators are evaluated, including Bollinger Bands, moving averages, and OBV. Dynamic parameter tuning occurred using Bayesian Optimization, the PCA technique was used, and attribute selection was performed, with the others evaluated using MAE and MAPE metrics. The RAJ study forecasts the S & P index using Deep Learning, with data from 2023 to 2025. The technical indicators used include Bollinger Bands, Moving Averages such as SMA and EMA, RSI, MACD, and ATR. The evaluation metrics employed are MSE, RMSE, MAE, MAPE, and Directional Accuracy. Both studies employ technical indicators used in this article, such as Bollinger Bands, SMA, and EMA, in addition to feature selection, and thus demonstrated performance when used in predictive models.

The TANRIKULU study compares the use of continuous data versus trend data on Bitcoin price movements, over a period of 2014 to 2021. Technical indicators include Bollinger Bands, SMA, EMA, Momentum, Stochastic, among others, as well as sentiment data. Various machine learning models, such as Random Forest, and neural networks, such as Artificial Neural Network, are used. Their evaluation metrics include Accuracies, F-start, AUC, and t-test. In the study discussed, the relative performances between Random Forest and Neural Networks showed the superiority of neural networks, obtaining better results, which can be compared in this article, which uses Random Forest and MLP, in addition to similar technical indicators.